

Bakery Data Analysis Report Using Data Analytics & Visualization

1. Introduction

This project focuses on analyzing bakery transaction data to understand customer behavior and improve business performance.

In today's data-driven world, businesses rely on insights extracted from data to optimize operations and increase profitability.

The main objective is to identify patterns in sales, peak time periods, and popular products.

2. Problem Statement

The bakery aims to understand customer buying behavior and improve sales performance.

The key problems include identifying peak sales times, top-selling products, and improving low-sales periods.

This project addresses these problems using data analysis and visualization.

3. Dataset Description

The dataset contains transaction-level information.

Each row represents a single item purchased in a transaction.

Columns include Transaction Number, Items, DateTime, Daypart, and DayType.

TransactionNo	Items	DateTime	Daypart	DayType
1	Bread	10/30/2016 9:58	Morning	Weekend
2	Scandinavian	10/30/2016 10:05	Morning	Weekend
2	Scandinavian	10/30/2016 10:05	Morning	Weekend
3	Hot chocolate	10/30/2016 10:07	Morning	Weekend
3	Jam	10/30/2016 10:07	Morning	Weekend
3	Cookies	10/30/2016 10:07	Morning	Weekend
4	Muffin	10/30/2016 10:08	Morning	Weekend
5	Coffee	10/30/2016 10:13	Morning	Weekend
5	Pastry	10/30/2016 10:13	Morning	Weekend
5	Bread	10/30/2016 10:13	Morning	Weekend
6	Medialuna	10/30/2016 10:16	Morning	Weekend
6	Pastry	10/30/2016 10:16	Morning	Weekend
6	Muffin	10/30/2016 10:16	Morning	Weekend
7	Medialuna	10/30/2016 10:19	Morning	Weekend
7	Pastry	10/30/2016 10:19	Morning	Weekend
7	Coffee	10/30/2016 10:19	Morning	Weekend
7	Tea	10/30/2016 10:19	Morning	Weekend
8	Pastry	10/30/2016 10:20	Morning	Weekend
8	Bread	10/30/2016 10:20	Morning	Weekend
9	Bread	10/30/2016 10:21	Morning	Weekend
9	Muffin	10/30/2016 10:21	Morning	Weekend
10	Scandinavian	10/30/2016 10:25	Morning	Weekend
10	Medialuna	10/30/2016 10:25	Morning	Weekend

4. Data Preprocessing and Cleaning

The dataset was cleaned to ensure consistency and accuracy.

DateTime column was converted into proper datetime format.

New features such as date, time, month, and hour were extracted.

Text fields were standardized by removing spaces and converting to lowercase.

A	B	C	D	E	F	G	H	I	J
TransactionNo	Items	DateTime	Daypart	DayType	Date	Time	Year	Month	Hour
1	bread	10/30/2016 9:58	morning	weekend	10/30/2016	9:58:11	2016	10	9
2	scandinavian	10/30/2016 10:05	morning	weekend	10/30/2016	10:05:34	2016	10	10
2	scandinavian	10/30/2016 10:05	morning	weekend	10/30/2016	10:05:34	2016	10	10
3	hot chocolate	10/30/2016 10:07	morning	weekend	10/30/2016	10:07:57	2016	10	10
3	jam	10/30/2016 10:07	morning	weekend	10/30/2016	10:07:57	2016	10	10
3	cookies	10/30/2016 10:07	morning	weekend	10/30/2016	10:07:57	2016	10	10
4	muffin	10/30/2016 10:08	morning	weekend	10/30/2016	10:08:41	2016	10	10
5	coffee	10/30/2016 10:13	morning	weekend	10/30/2016	10:13:03	2016	10	10
5	pastry	10/30/2016 10:13	morning	weekend	10/30/2016	10:13:03	2016	10	10
5	bread	10/30/2016 10:13	morning	weekend	10/30/2016	10:13:03	2016	10	10
6	medialuna	10/30/2016 10:16	morning	weekend	10/30/2016	10:16:55	2016	10	10
6	pastry	10/30/2016 10:16	morning	weekend	10/30/2016	10:16:55	2016	10	10
6	muffin	10/30/2016 10:16	morning	weekend	10/30/2016	10:16:55	2016	10	10
7	medialuna	10/30/2016 10:19	morning	weekend	10/30/2016	10:19:12	2016	10	10
7	pastry	10/30/2016 10:19	morning	weekend	10/30/2016	10:19:12	2016	10	10
7	coffee	10/30/2016 10:19	morning	weekend	10/30/2016	10:19:12	2016	10	10
7	tea	10/30/2016 10:19	morning	weekend	10/30/2016	10:19:12	2016	10	10
8	pastry	10/30/2016 10:20	morning	weekend	10/30/2016	10:20:51	2016	10	10
8	bread	10/30/2016 10:20	morning	weekend	10/30/2016	10:20:51	2016	10	10
9	bread	10/30/2016 10:21	morning	weekend	10/30/2016	10:21:59	2016	10	10
9	muffin	10/30/2016 10:21	morning	weekend	10/30/2016	10:21:59	2016	10	10
10	scandinavian	10/30/2016 10:25	morning	weekend	10/30/2016	10:25:58	2016	10	10
10	medialuna	10/30/2016 10:25	morning	weekend	10/30/2016	10:25:58	2016	10	10

5. Feature Engineering

New features were created to enhance analysis.

Hour helps identify peak sales time.

Month helps analyze trends over time.

Date helps track daily patterns.

6. Key Performance Indicators (KPIs)

Total Sales: 21,000 items sold.

Total Orders: 9,465 transactions.

Average Items per Order: 2.17.

These KPIs indicate that customers often purchase multiple items.

9465 Total Orders	21K Total Sales	2.17 Avg Items per Order
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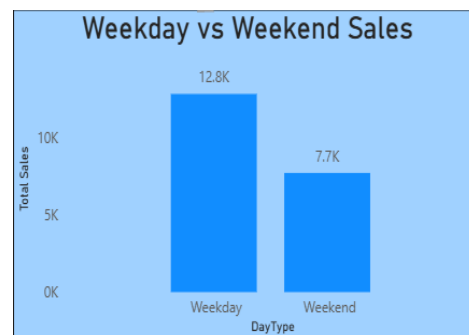
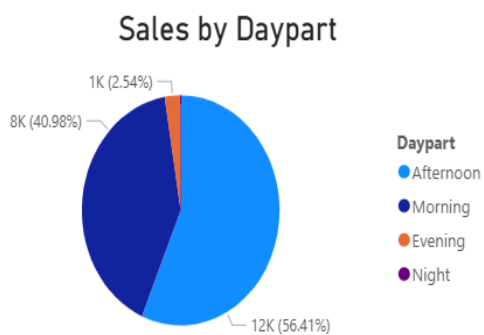
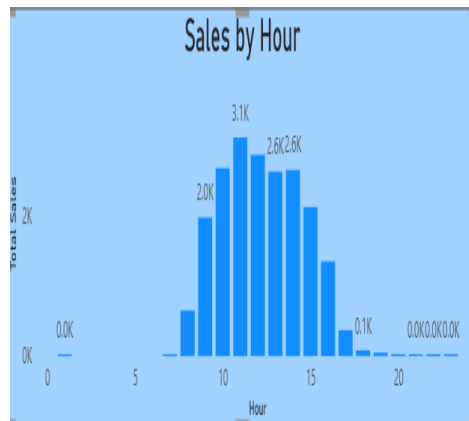
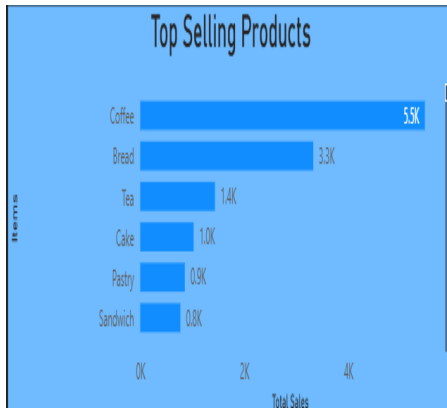
7. Exploratory Data Analysis (EDA)

EDA was performed to uncover patterns in sales and customer behavior.

Top-selling items include coffee, bread, and tea.

Peak sales occur during morning hours (8–11 AM).

Weekday sales are higher than weekend sales.

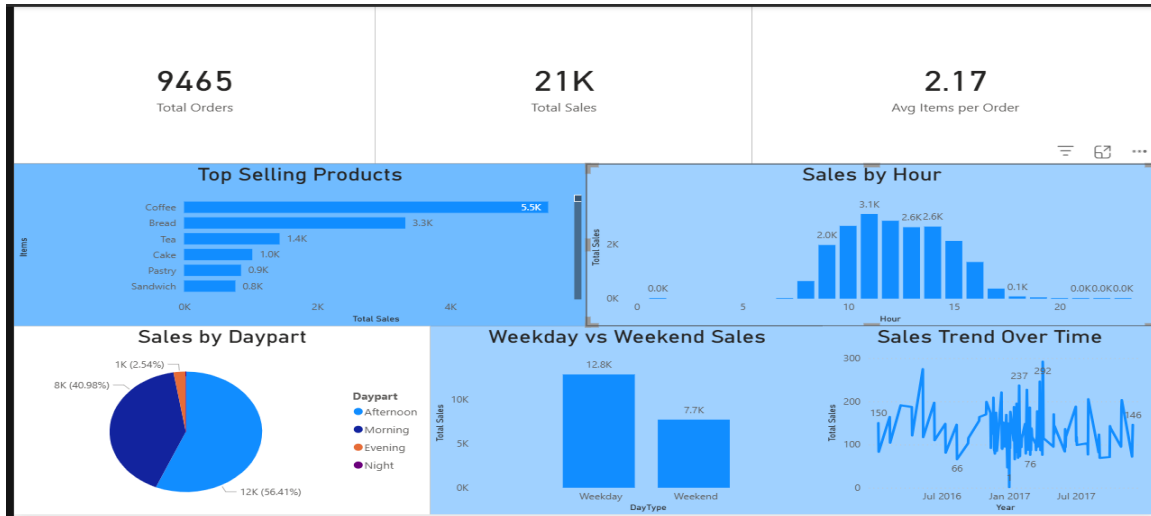


8. Dashboard and Visualization

An interactive dashboard was created using Power BI.

It includes KPI cards, top products, sales trends, and time analysis.

The dashboard helps in understanding data quickly and supports decision-making.



9. Hypothesis Testing

Objective: To test whether morning sales are higher than other time periods.

H0: Morning sales are equal to other time periods.

H1: Morning sales are higher than other time periods.

Result: Morning sales are significantly higher.

Conclusion: Reject null hypothesis.

10. Business Insights

Morning hours drive the majority of sales.

A few products contribute to most of the revenue.

Customers tend to purchase multiple items per order.

Sales patterns remain consistent across time.

11. Recommendations

Introduce combo offers such as coffee and bread.

Increase staffing during peak morning hours.

Provide discounts during evening to boost sales.

Focus inventory on high-demand products.

12. Conclusion

This project successfully analyzed bakery sales data and identified key patterns.

The insights can help improve business performance and decision-making.

Data analytics and visualization play an important role in modern businesses.